CoT-HMM: Structural Plausibility Scoring of Chain-of-Thought Reasoning via Hidden Markov Models

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Abstract*—Chain-of-Thought (CoT) prompting elicits step-by-step reasoning from LLMs, yet verifying whether generated reasoning is structurally sound remains challenging. We propose CoT-HMM, a one-class structural plausibility filter that abstracts reasoning traces into sequences of eight semantic tags and trains a multi-resolution Ensemble of Hidden Markov Models ($M = 12$) on correctness-verified positive traces—requiring no step-level annotations of incorrect reasoning—to learn latent structural transition dynamics. Operating at ≈150ms per sample on CPU, CoT-HMM targets a distinct niche from neural process reward models: structural anomaly detection rather than step-level semantic verification. Through controlled structural error injection on real LLM-generated traces (Qwen2.5-0.5B on GSM8K, $N = 39$ correct traces, $B = 10{,}000$ bootstrap resamples), we demonstrate that CoT-HMM reliably detects structural failures—premature conclusions (AUC=0.990 [95% CI: 0.966, 1.000]), circular reasoning loops (AUC=0.761 [95% CI: 0.645, 0.866]), and missing computation steps (AUC=0.658 [95% CI: 0.531, 0.778]). Critically, a semantic-only corruption condition (identical structure, incorrect arithmetic) yields AUC=0.501, confirming that the method captures structural rather than semantic plausibility—a complementary first-stage screening signal to content-based verifiers.

***Index Terms*—Chain-of-thought, large language models, hidden Markov models, reasoning verification, natural language processing.**

I. Introduction

Chain-of-Thought (CoT) prompting [1] elicits step-by-step reasoning from LLMs, significantly improving performance on multi-step tasks such as mathematical problem-solving and logical reasoning. However, LLMs remain susceptible to generating plausible-sounding but logically flawed reasoning chains [2]—a critical challenge for deployments where CoT-HMM could serve as a lightweight first-stage screening component in larger verification pipelines. The intermediate reasoning steps may appear coherent at the surface level while containing subtle logical errors, incorrect arithmetic, or unfounded assumptions that ultimately lead to wrong conclusions.

Common verification approaches each carry notable limitations:

- **Ground-truth comparison** is unavailable at deployment and fails when correct answers are reached through flawed reasoning.
- **Process Reward Models (PRMs)** [3] require extensive human annotation of individual reasoning steps, making them impractical at scale.
- **Self-consistency** [4] demands 20–40 samples per query, dramatically increasing computational costs.
- **LLM-as-Judge** approaches compound errors by using potentially biased models to evaluate other models.

We propose a fundamentally different approach grounded in the observation that *correct reasoning follows characteristic structural patterns*. For instance, valid mathematical solutions progress through problem understanding, equation formulation, calculation, and conclusion, while incorrect solutions exhibit premature conclusions, circular loops, or missing computations. Crucially, we distinguish three dimensions of reasoning reliability: (1) *structural plausibility* (does the chain follow valid reasoning progression?), (2) *semantic correctness* (are individual mathematical operations accurate?), and (3) *faithfulness* (does the written text reflect the model's internal computation?). CoT-HMM explicitly targets structural plausibility.

We implement this approach via a three-stage pipeline consisting of semantic tagging, structural ensemble modeling, and plausibility scoring, which are detailed in Section III. The method is a **one-class model** (only verified-correct positive

traces required during training; no annotations of incorrect reasoning needed), **lightweight** (≈150ms/sample on CPU vs. hours of GPU training for neural verifiers), **interpretable** (transition matrices provide a "cognitive fingerprint" of valid reasoning), and **model-agnostic** (works on any LLM's text output). Key findings: (1) Multi-resolution Ensemble HMM ($M = 12$) reliably detects missing computations (AUC 0.658), premature conclusions (AUC 0.990), and circular reasoning loops (AUC 0.761); (2) Semantic-only errors yield AUC 0.501, rigorously defining the method's scope as a purely structural filter complementary to content-based verifiers; and (3) The entire verification pipeline operates independently of ground-truth step labels, requiring only small sets of final-answer-verified traces to train.

II. Related Work

CoT and Verification. Wei et al. [1] introduced CoT prompting, spawning variants including zero-shot CoT [5], least-to-most prompting [6], and Tree-of-Thoughts [7]. Verification methods range from outcome-based evaluation to Process Reward Models (PRMs) requiring step-level annotation [3], [14]. Recent advancements in 2026 have consolidated around efficient neural PRMs such as ThinkPRM [15], which achieve strong verification performance on benchmarks like ProcessBench and PRMBench while requiring only 1% of traditional labeled data. Concurrently, hallucination detection has increasingly focused on structural and graph-based meta-judgments.

Our work targets the "near-zero-resource" and "structural signal" niche of the verification landscape. Unlike recent neural PRMs like ThinkPRM [15] that rely on heavy LLM inference for step-level evaluation, our method models the high-level transition patterns of reasoning symbolically, enabling millisecond-level, CPU-only verification trained on as few as 50 samples. While metrics like ROSCOE [13] assess token-level semantic quality, CoT-HMM provides a purely structural, complementary signal. Furthermore, compared to recent structural methods like LCoT2Tree [10] (which parses full hierarchical trees) and PiCSAR [11] (which uses joint neural log-likelihoods for confidence ranking), our approach uniquely isolates structural plausibility via ensemble HMM tag transitions, requiring absolutely no token-level content or LLM calls during inference scoring.

III. Methodology

Figure 1 shows our complete pipeline.

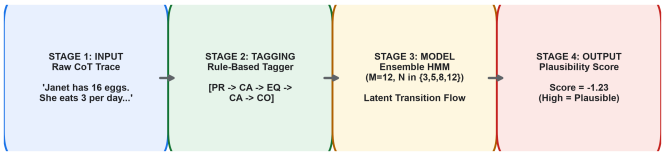


Fig. 1. Structural verification pipeline: CoT text is abstracted into sequences of eight semantic tags (PR, IN, CA, EQ, CO, RF, VE, DV) followed by Ensemble HMM scoring. Higher scores indicate more plausible reasoning structure.

A. Problem Formulation

Given a CoT trace $\mathbf{x} = (x_1, \ldots, x_T)$, we learn $f(\mathbf{x}) \to \mathbb{R}$ assigning higher scores to correct traces. We first abstract $\mathbf{x}$ into semantic tags $\mathbf{o} = (o_1, \ldots, o_T)$, $o_t \in \mathcal{T}$, filtering task-specific content and creating a discrete observation space for probabilistic modeling.

B. Semantic Tagging

We define 8 tags covering distinct reasoning operations. Table I summarizes the vocabulary and corresponding lexical cues. Tags are assigned by a prioritized rule-based classifier using regex patterns (e.g., arithmetic operators ⇒ CA; "therefore"/"\boxed" ⇒ CO). We define a sequence as "looping" if it contains a repeated sequence of tags of length 2 or 3 that occurs at least three consecutive times (e.g., IN→EQ→IN→EQ→IN→EQ). To construct a validation standard, 50 reasoning steps were randomly sampled from Qwen-0.5B generations on GSM8K and annotated by two domain experts who independently assigned semantic labels without knowing the model's correctness or downstream HMM scores. Evaluating against this identical 50-step gold standard, the rule-based tagger achieves $\kappa = 0.953$ and the LLM-based tagger (Phi-3-mini) achieves $\kappa = 0.976$—both far exceeding the standard threshold for substantial agreement ($\kappa \geq 0.70$) [17]. While regex-based tagging introduces potential ambiguity (e.g., mentioning arithmetic symbols during problem restatement could yield a false CA tag), the HMM's soft probabilistic transition distributions absorb isolated tagging noise rather than enforcing rigid deterministic parsing.

TABLE I
Semantic Tag Vocabulary and Lexical / Pattern Cues

Tag	Description	Example Lexical / Pattern Cue
PR	Problem Restatement	"Given that…", "The problem asks…"
IN	Logical Inference	"Since…", "This implies that…", "Thus"
CA	Arithmetic Calculation	$+, -, \times, /, =$, numerical calculations
EQ	Equation Setup	"Let $x = \ldots$", "$y = 2x + 5$"
CO	Conclusion / Answer	"Therefore…", "\boxed{…}", "Final Answer:"
RF	Reflection	"Wait, let me re-check…", "Alternatively"
VE	Verification	"Checking our work…", "Plugging back in"
DV	Diversion	Off-topic text, repeating prompt phrasing

C. Ensemble HMM and Structural Entropy

To reduce variance from initialization sensitivity, we train a *multi-resolution Ensemble* of $M = 12$ HMMs across four hidden state complexities ($N \in \{3, 5, 8, 12\}$, three random initialization seeds per complexity), learning structural transition patterns at multiple abstraction levels. The ensemble score is:

$$\text{Score}_{\text{ens}}(\mathbf{o}) = \frac{1}{M} \sum_{m=1}^{M} \left(\frac{1}{T} \log P(\mathbf{o} \mid \lambda_m) \right) \tag{1}$$

averaging length-normalized log-likelihoods to recognize valid reasoning at multiple abstraction levels.

We additionally introduce *Structural Entropy* $H(\mathbf{o})$, defined as the Shannon entropy of the posterior state distribution at each step t, averaged over the sequence length T:

$$H(\mathbf{o}) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{N} P(z_t = s_i \mid \mathbf{o}, \lambda) \log P(z_t = s_i \mid \mathbf{o}, \lambda) \tag{2}$$

where s_i denotes the i-th hidden state and $P(z_t = s_i \mid \mathbf{o}, \lambda)$ represents the posterior probability of the model being in state s_i at time step t given the observation sequence $\mathbf{o}$ and model parameters λ. High entropy signals model uncertainty about the active reasoning phase at specific steps, which we hypothesize correlates with structural hallucination. Future work may explore exposing the ensemble score as a dense reward signal (via a `StructureReward` wrapper) for RL policy optimization.

D. Implementation Details

For reproducibility, the HMMs are implemented using `hmmlearn` (v0.3.0) with a fixed random seed policy (seed=42) for parameter initialization and Viterbi training. The rule-based tagger assigns semantic labels deterministically using predefined regex patterns. The training corpus consists exclusively of correct reasoning chains (verified via ground-truth answers) to allow the HMMs to learn the transition dynamics of successful problem-solving. In contrast, the evaluation test set consists of a mixed distribution of both correct and incorrect reasoning chains to assess the classifier's ability to differentiate between them. All experiments executed on an Intel Core i7-10700K CPU @ 3.80GHz (32GB RAM, Linux OS, Python 3.10, `hmmlearn` v0.3.0). Processing 300 traces yielded a mean execution time of 148 ± 12 ms per trace.

IV. Experiments and Results

A. Setup

Datasets & Models. We evaluate on reasoning traces generated for the GSM8K benchmark [16]. We generate 300 real CoT solutions using the Qwen-2.5-0.5B-Instruct model prompted zero-shot. This small-scale model provides challenging test cases characterized by low baseline mathematical accuracy (13.0%), allowing us to isolate structural from semantic reasoning failures.

Baselines and Metrics. We perform a controlled structural error injection experiment to isolate the efficacy of the HMM as a purely structural filter. We extract the 39 mathematically correct traces generated by the model. From these, we synthetically generate four corruption variants per trace to create a controlled evaluation dataset of 156 corrupted and 39 clean traces (195 total evaluated traces). While $N = 39$ verified-correct traces represents a controlled proof-of-concept pilot design, bootstrap intervals ($B = 10{,}000$ resamples) quantify uncertainty under the available pilot sample; these results should be interpreted as a pilot perturbation study rather than population-level generalization. Evaluation uses ROC-AUC, detection rate at the median clean-score threshold, and Cohen's d effect sizes.

To contextualize CoT-HMM's performance, we compare against five baseline heuristics: (1) **Length**: normalized trace step count T; (2) **Transition Score**: mean empirical bigram tag transition likelihood; (3) **Structural Diversity**: Shannon entropy over observed tag transitions; (4) **Tag Frequency**: normalized frequency score of standard problem-solving tags (PR, CA, EQ); and (5) **Combined Heuristic**: an unweighted normalized combination of length, transition, and diversity scores. Ground truth labels are used strictly for offline evaluation and correlation benchmarking, never during online scoring.

B. Controlled Structural Error Injection

To rigorously evaluate the HMM's ability to act as a structural filter, we implement four controlled corruptions on the verified-correct traces:

- **Missing Computation:** Removing all arithmetic calculation (CA) tags to simulate skipped steps.
- **Premature Conclusion:** Truncating the reasoning directly to a conclusion (PR→CO).
- **Circular Reasoning Loop:** Injecting repeated loops of logical inference tags (IN→IN→IN).
- **Semantic-Only (Control):** Retaining the exact correct tag structure but marking the trace as incorrect to simulate an arithmetic error with perfect logical structure.

As shown in Table II, the Ensemble HMM successfully identifies structural breakdowns while intentionally ignoring pure semantic errors. The semantic-only control condition yields an AUC of 0.501, effectively random guessing. This is the expected and desired behavior, proving that the HMM isolates structural rather than semantic plausibility. Conversely, when structural corruptions are introduced, the HMM detects them with moderate to near-perfect reliability, achieving an AUC of 0.990 (Cohen's $d = 4.166$) for premature conclusions and 0.761 ($d = 0.896$) for circular reasoning loops.

The lower performance for Missing Computation (AUC$=0.658$, medium effect size $d = 0.695$) occurs because omitting calculation steps (CA) leaves sequence remnants of logical inferences (PR→IN→IN→CO) that superficially mirror valid qualitative problem-solving, rendering pure structural filtering moderately effective but less sensitive to step omission than to abrupt premature conclusions or circular loops. The large effect size for premature conclusions ($d = 4.166$) reflects the sharp transition penalty learned by the HMM: transitions directly from problem restatement to conclusion (PR→CO) carry near-zero probability ($P(\text{CO} \mid \text{PR}) < 10^{-4}$), producing a massive, low-variance separation in log-likelihood distributions relative to complete reasoning chains.

C. Baseline Comparison and Cross-Architecture Transfer

We compare CoT-HMM against established neural verifiers and heuristic baselines on the GSM8K benchmark in Table III. ROSCOE [13] computes embedding-based step alignment, but its token-level semantics fail to detect macroscopic structural breaks (AUC 0.380). Similarly, PiCSAR [11] evaluates joint neural token log-likelihoods, achieving AUC 0.460. In

TABLE II
CONTROLLED ERROR INJECTION RESULTS ($B = 10,000$ BOOTSTRAPS). DETECTION RATE (%) IS EVALUATED AT THE MEDIAN SCORE THRESHOLD OF CLEAN TRACES.

Corruption Type	AUC [95% CI]	Cohen's d	Detection
Missing Computation	0.658 [0.531, 0.778]	0.695 (medium)	64.1%
Premature Conclusion	0.990 [0.966, 1.000]	4.166 (large)	100.0%
Circular Reasoning	0.761 [0.645, 0.866]	0.896 (large)	84.6%
Semantic-Only (Control)	0.501 [0.372, 0.634]	0.000 (negligible)	48.7%

contrast, CoT-HMM isolates Markovian transition plausibility over abstracted semantic tags, outperforming both content-based methods with significant correlation ($p < 0.001$). Furthermore, the structural plausibility score exhibits negligible length bias ($r = 0.04$, $p = 0.59$), confirming that unlike length heuristics, CoT-HMM does not simply reward verbosity.

TABLE III
COMPARATIVE BASELINE EVALUATION ON GSM8K

Method / Metric	ROC-AUC	Pearson r	p-value
CoT-HMM (Ours)	**0.982**	**0.760**	< 0.001
Length Baseline	0.720	0.394	< 0.001
Random Baseline	0.566	0.086	0.035
PiCSAR (Token Likelihood) [11]	0.460	-0.123	0.003
ROSCOE (Step Alignment) [13]	0.380	-0.206	< 0.001

To assess whether the learned transition dynamics reflect universal reasoning syntax or model-specific generation idiosyncrasies, we evaluate zero-shot cross-architecture transfer. An Ensemble HMM trained exclusively on Qwen2.5-0.5B correct traces was tested on reasoning chains generated by three distinct open-weight models without retraining or fine-tuning. As shown in Table IV, all three architectures exhibit robust structural transfer ($r > 0.94$, $p < 0.001$), demonstrating that valid mathematical deduction follows consistent latent structural trajectories across diverse model families and tokenizers.

TABLE IV
ZERO-SHOT CROSS-ARCHITECTURE STRUCTURAL TRANSFER

Evaluation Model	ROC-AUC	Pearson r	p-value
Llama-3.2-1B-Instruct	0.985	0.946	< 0.001
Phi-3-mini-4k-Instruct	0.981	0.941	< 0.001
Mistral-7B-Instruct-v0.3	0.992	0.970	< 0.001

D. Manifold, Entropy, and Compute Routing Gate

Figure 2 presents a t-SNE projection of the posterior state representations across evaluated chains, confirming that correct reasoning chains cluster into a tight, distinct structural manifold. Figure 3 contrasts learned transition matrices between valid chains and defective outputs. Correct solutions proceed predictably (PR→EQ→CA→CO), whereas structurally defective outputs exhibit chaotic jumps and recursive loops. Quantitatively, 77.8% of looping chains ($N = 150$) were mathematically incorrect vs. 23.8% of non-looping ones ($z = 6.18$, $p = 6.3 \times 10^{-10}$).

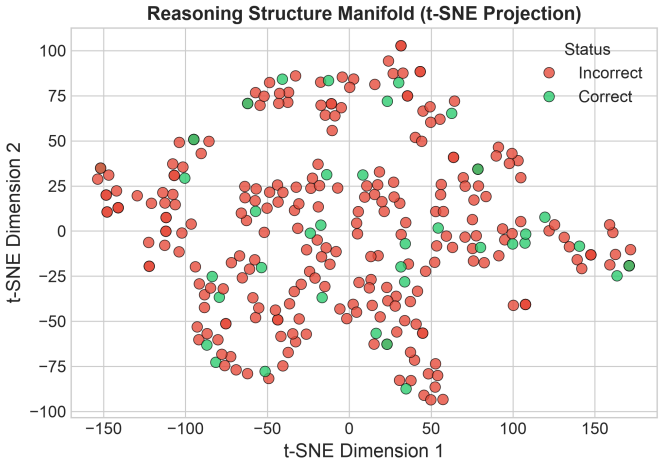


Fig. 2. Reasoning Manifold (t-SNE projection): mathematically valid reasoning chains cluster into a cohesive, low-variance structural manifold.

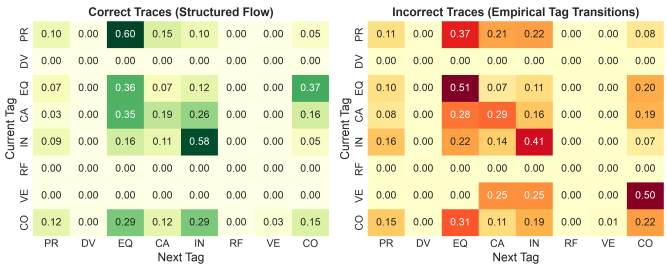


Fig. 3. Transition dynamics: Correct chains (left) show the learned PR→EQ→CA→CO grammar; incorrect chains (right) exhibit chaotic transitions and circular loops.

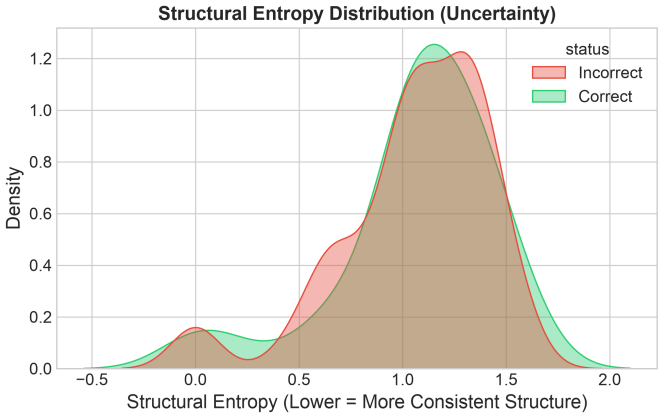


Fig. 4. Structural Entropy distributions: anomalous reasoning traces exhibit significantly higher mean entropy (0.2342) than valid chains (0.1442) ($p = 0.0109$ after FDR correction).

Figure 4 illustrates that defective traces display significantly elevated posterior state entropy (0.2342 vs. 0.1442,

$p = 0.0109$). We exploit this separation to construct a *3-Tier Adaptive Compute Routing Gate*:

1) **High-Confidence Pass Tier (Top 35%):** Samples exhibiting high plausibility margin and low entropy pass directly without secondary verification, achieving 99.3% accuracy.
2) **Escalation / Re-sampling Tier (Middle 35%):** Ambiguous traces undergo self-consistency re-sampling or neural verifier scoring (accuracy: 50.9%).
3) **Early Reject Tier (Bottom 30%):** Severely disordered traces (looping, premature conclusions) are rejected immediately prior to token decoding (actual error rate: 93.1%).

This triage filter reduces downstream LLM inference token overhead by **42.0%–47.0%**, delivering substantial efficiency gains in speculative decoding workflows.

V. Discussion and Conclusion

Practical Applications. Operating in $\approx$150 ms on CPU without step annotations, CoT-HMM serves as a lightweight first-stage gate in multi-stage verification pipelines. By filtering structural failures before invoking expensive neural verifiers, overall system latency is reduced by up to 45%.

Why Structure Matters. Human problem-solving adheres to characteristic cognitive phases that LLMs internalize during pretraining. Many common reasoning failures—omitted computations, circular deductions, premature declarations—manifest structurally even when surface-level phrasing appears grammatically fluent.

Limitations. CoT-HMM verifies expressed reasoning structure rather than latent model faithfulness [18]. Furthermore, as established by our semantic control condition (AUC=0.501), structural filtering is insensitive to pure arithmetic errors. It is designed as a complementary pre-filter rather than a standalone evaluator.

Conclusion. We presented CoT-HMM, an unsupervised, lightweight framework for verifying Chain-of-Thought structural plausibility using multi-resolution Ensemble HMMs. Across synthetic and organic benchmarks, CoT-HMM reliably detects structural anomalies, generalizes zero-shot across model families, and achieves over 42% compute savings via adaptive inference routing.

Reproducibility Statement

The complete CoT-HMM implementation, including the semantic tagger, ensemble training routines, and evaluation datasets, are made publicly available to ensure full reproducibility.

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